

MarineGuard AI — Investigation Report

SYNTHETIC DEMONSTRATION DATA — not real-world observations.

1. Incident information

Field	Value
Incident ID	MG-2026-DEMO-001
Title	Synthetic Arabian Sea oil-slick investigation
Location	Arabian Sea, west of Mumbai (synthetic scene)
Observation time	2026-03-14T06:30:00Z
Provenance	SYNTHETIC DEMONSTRATION DATA

2. Satellite observation

SAR scene associated with this incident. Detection mode: **Prototype/Demo Detection**. No trained weights were loaded. This is a prototype/demo detector, not a real ML prediction.

3. Spill detection

Field	Value
Mode	Prototype/Demo Detection
Confidence	68.0%
Status	Completed

4. Spill geometry

Field	Value
Area	45.4 km²
Length	17.77 km
Width	3.41 km
Perimeter	39.59 km
Centroid	19.10398, 71.83161
Shape	elongated

5. Probable origin

Field	Value
Latitude	19.11449
Longitude	71.76011
Estimated origin time	2026-03-14T02:00:00Z
Window	2026-03-14T00:30:00Z → 2026-03-14T03:30:00Z
Uncertainty radius	5.45 km

This is a simplified prototype drift model (ocean current + 3% wind factor). It is not a hydrodynamic or ocean-circulation model.

6. Hindcast result

Backward trajectory points: 13. Model: simplified_prototype_drift.

7. Forecast

Lead time	Latitude	Longitude
+0 h	19.10398	71.83161
+1 h	19.10295	71.84941
+2 h	19.10201	71.86765
+3 h	19.10149	71.88579
+6 h	19.10178	71.93722
+12 h	19.10799	72.03086

Forecast uses the same simplified current + wind-factor model as hindcast. Uncertainty grows with lead time.

8. AIS traffic analysis

Field	Value
Total vessels	7
Spatial candidates	7
Temporal candidates	7
Final candidates	7

9. Candidate vessel ranking

Rank	MMSI	Score	Priority
1	419000111	86.6	HIGH
2	419000222	80.2	HIGH
3	419000444	74.9	MEDIUM
4	419000333	69.5	MEDIUM
5	419000777	66	MEDIUM
6	419000666	62.6	MEDIUM
7	419000555	59.5	MEDIUM

10. Evidence

MT SYNTHETIC STAR (419000111) — Highest-priority candidate

- Closest approach 5.1 km from probable origin.
- Present during estimated spill window.
- Mean COG 85.7° vs origin bearing (Δ 39.1°).
- Slow/stopped behaviour near origin.

MV COASTAL PEARL (419000222) — Probable source vessel candidate

- Closest approach 3.2 km from probable origin.
- Present during estimated spill window.
- Mean COG 97.2° vs origin bearing (Δ 4.8°).
- Typical speed 12.0 kn.

MV NORTHBOUND (419000444) — Probable source vessel candidate

- Closest approach 5.8 km from probable origin.
- Present during estimated spill window.
- Mean COG 181.2° vs origin bearing (Δ 33.4°).
- Typical speed 14.54 kn.

FV WESTERN NET (419000333) — Probable source vessel candidate

- Closest approach 9.9 km from probable origin.
- Present during estimated spill window.
- Mean COG 17.2° vs origin bearing (Δ 41.2°).
- Typical speed 8.0 kn.

SV LOCAL HARBOR (419000777) — Probable source vessel candidate

- Closest approach 15.2 km from probable origin.
- Present during estimated spill window.
- Mean COG 315.8° vs origin bearing (Δ 5.5°).
- Typical speed 6.67 kn.

11. Confidence

Detection confidence 68.0%. Drift uncertainty radius 5.45 km. Forecast corridor width 8.6 km.

12. Data gaps

- Satellite detection may include false positives, especially in prototype/demo mode.
- AIS coverage may be gapped, delayed, or spoofed.
- Ocean wind/current fields are sparse demonstration vectors, not a circulation model.
- 419000333: AIS gap of 1.33 hours.
- 419000666: AIS gap of 4.17 hours.
- 419000555: AIS gap of 2.0 hours.

13. Responsible-AI disclaimer

This system prioritises vessels for investigation and does not establish definitive legal responsibility. The highest-ranked vessel is a highest-priority candidate only. Results require human validation and must not be used as the sole basis for legal enforcement.